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Grade	**51.00** out of 66.00 (**77**%)

Question 1

Partially correct

Mark 1.00 out of 3.00

Theme: ICD-11 Criteria
Lead-in: Identify the criterion that most differentiates each of these disorders in ICD-11.

1. Selective mutism > 6 months duration ✖

2. Social anxiety disorder Fear of being negatively evaluated ✔

3. Separation anxiety disorder Typical onset before age 5 years ✖

Explanation: Selective mutism is characterised by a failure to speak in specific situations, whereas in social anxiety disorder, fear/anxiety results in an avoidance of multiple social contexts. Unlike social anxiety disorder, children with selective mutism are likely to display these difficulties at an earlier age of onset (before age of 5 years).
Social anxiety disorder is less common in young children under the age of 10, with occurrence increasing significantly during adolescence. In social anxiety disorder, avoidance of social situations is in response to fear/anxiety about being negatively evaluated by others rather than concerns about being separated from key attachment figures, as in separation anxiety disorder. A differentiating feature of separation anxiety disorder is recurrent nightmares about separation.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization, 2020)

Correct Answers:
1) Selective mutism - Typical onset before age 5 years
2) Social anxiety disorder - Fear of being negatively evaluated
3) Separation anxiety disorder - Nightmares about separation

Question 2

Correct

Mark 3.00 out of 3.00

Theme: Concepts of Piaget
Lead-in: Match each of the following concepts with its Piagetian term.

1. Compensation for external disturbance that occurs when previous cognitive structures are inadequate — Equilibration ✔

2. A type of thinking that moves from the general to the particular — Deductive reasoning ✔

3. Having a limited point of view and being unable to take the role of another person — Egocentric thinking ✔

Explanation: When an individual sees that previous cognitive structures are inadequate, the old cognitive structure is disturbed. This disruption leads to a search for more adequate cognitive structures. The compensation for external disturbance is what Piaget termed equilibration. New structures can only be constructed through a process of accommodation, enabling the individual to assimilate a more comprehensive array of data, a new perspective, or more complex information.
Deductive reasoning moves from the general to the particular and is a more complicated process than inductive reasoning, which moves from the particular to the general. In Piaget's stage of preoperational thought, children are considered to be egocentric; they see themselves as the centre of the universe, have a limited point of view, and cannot take the role of another person. Egocentric thinking prevents an understanding of another's point of view.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1007-1010.

Correct Answers:
1) Compensation for external disturbance that occurs when previous cognitive structures are inadequate - Equilibration
2) A type of thinking that moves from the general to the particular - Deductive reasoning
3) Having a limited point of view and being unable to take the role of another person - Egocentric thinking

Question 3

Correct

Mark 3.00 out of 3.00

Theme: Watson and Rayner's 'Little Albert' study
Lead-in: In the seminal 'Little Albert" study conducted by Watson and Rayner, young Albert was initially exposed to a rat, and the boy showed no fear. The next time Albert was exposed to the rat, Watson made a loud noise, which caused the child to cry. After repeatedly pairing the rat with the loud noise, Albert began to cry simply after being exposed to the rat (in the absence of the loud noise). In behavioural terms

1. What is the rat? Conditioned stimulus ✔

2. What is the loud noise? Unconditioned stimulus ✔

3. What is the crying in response to the rat? Conditioned response ✔

Explanation: Since the publication of Watson and Rayner's seminal 'Little Albert' study, fear conditioning has become one of the most widely used paradigms to study the acquisition, extinction, and return of fear. The core procedural elements of the fear conditioning paradigm are the pairing of an initially neutral stimulus, called the conditioned stimulus (CS), with an aversive unconditioned stimulus (US), resulting in the establishment of a conditioned response (CR) to the CS, even in the absence of the US. The 'Little Albert' study pioneered the psychological theory of behaviourism, which focuses on analysing behaviour in terms of observable environmental elements (such as CSs and USs) and their effects (observable CRs). The loud noise is the unconditioned stimulus, the rat is the conditioned stimulus, and the conditioned response is the crying in response to the rat.

Ref: Mertens G et al. A review on mental imagery in fear conditioning research 100 years since the 'Little Albert'study. Behaviour Research and Therapy. 2020 Mar 1;126:103556.

Correct Answers:
1) What is the rat? - Conditioned stimulus
2) What is the loud noise? - Unconditioned stimulus

3) What is the crying in response to the rat? - Conditioned response

Question **4**

Partially correct

Mark 2.00 out of 3.00

Theme: Stage-based theories and their proponents
Lead-in: Match each set of stages with its proponent's theory.

1. Protest -> Despair -> Detachment — Kohlberg's theory of moral development ✗

2. Shock/Denial -> Anger -> Bargaining -> Depression -> Acceptance — Kubler-Ross' theory of death and dying ✓

3. Numbing -> Yearning -> Disorganisation -> Reorganisation — Parkes & Bowlby's theory of grief ✓

Explanation: Bowlby described a predictable set and sequence of behaviour patterns in children who are separated from their mothers for extended periods: protest, despair, and detachment. Bowlby believed that the sequence involves ambivalent feelings toward the mother; the child both wants her and is angry with her for her desertion.
Kulber-Ross made a comprehensive and useful organisation of reactions to impending death. No established sequence applies to all individuals, but the five stages proposed by Kubler-Ross are: shock/denial, anger, bargaining, depression, and acceptance.
Parkes was inspired by Bowlby's attachment theory and collaborated with Bowlby in his research following a group of widows to track the course of normal adult grief. As a result, they described four phases of mourning: numbing, yearning, disorganisation, and reorganisation.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 847-848, 1013.
Ince HY, Balon R. Bowlby's Four Phases of Mourning Through Krzysztof Kieslowski's Blue. Acad Psychiatry 2024; (48) 89–92.

Correct Answers:
1) Protest -> Despair -> Detachment - Bowlby's theory of maternal deprivation
2) Shock/Denial -> Anger -> Bargaining -> Depression -> Acceptance - Kubler-Ross' theory of death and dying
3) Numbing -> Yearning -> Disorganisation -> Reorganisation - Parkes & Bowlby's theory of grief

Question 5

Correct

Mark 3.00 out of 3.00

Theme: Human Genes
Lead-in: Identify the gene that codes for each of the following:

1. IKr potassium channel hERG ✔

2. Alpha synuclein SNCA ✔

3. Tau MAPT ✔

Explanation: Modulation of the human Ether-a-go-go-Related Gene (hERG) channel, a crucial voltage-gated potassium channel in the repolarisation of action potentials in heart ventricular myocytes, has significant implications for cardiac electrophysiology.
Alpha-synuclein is a protein encoded by the SNCA gene. Missense mutations and gene duplications in the SNCA gene cause hereditary Parkinson's disease.
Microtubule Associated Protein Tau (MAPT) codes for tau, a microtubule-associated protein that plays a role in causing common neurodegenerative diseases and rare inherited frontotemporal lobar degenerations.

Ref: Kawahata I et al. Pathogenic impact of a-synuclein phosphorylation and its kinases in a-synucleinopathies. International journal of molecular sciences. 2022 Jun 1;23(11):6216.
Furutani K. Facilitation of hERG Activation by Its Blocker: A Mechanism to Reduce Drug-Induced Proarrhythmic Risk. International Journal of Molecular Sciences. 2023 Nov 13;24(22):16261.
Callari M et al. Cancer-specific association between Tau (MAPT) and cellular pathways, clinical outcome, and drug response. Scientific Data. 2023 Sep 20;10(1):637.

Correct Answers:
1) IKr potassium channel - hERG
2) Alpha synuclein - SNCA

3) Tau - MAPT

Question 6

Correct

Mark 3.00 out of 3.00

Theme: Classic syndromes and characteristic features of neuronal autoantibodies
Lead-in: Choose the most likely target antigen for each clinical presentation of autoimmune encephalitis.

1. Young females, acute onset, complex movement disorder NMDAR ✔

2. Elderly, focal seizures including facio-brachial dystonic seizures, hyponatremia LGI1 ✔

3. Young patients, optic neuritis, transverse myelitis MOG ✔

Explanation: NMDAR-antibody encephalitis most often occurs in young females. Characteristic features include acute onset, mixed psychopathology, complex movement disorder, often with hyperkinetic-catatonic fluctuations, and dysautonomia. Frequently, an ovarian teratoma is discovered.

LGI1-antibody encephalitis most often occurs in elderly males and can present insidiously. Characteristic features include frequent focal seizures (including facial-brachial dystonic seizures), amnesia, and serum hyponatraemia.
MOG-antibody encephalitis occurs most often in young patients. It typically presents similar to acute disseminated encephalomyelitis (ADEM) and can be accompanied by optic neuritis and/or transverse myelitis.

Ref: Uy CE et al. Autoimmune encephalitis: clinical spectrum and management. Practical neurology. 2021 Oct 1;21(5):412-23.

Correct Answers:
1) Young females, acute onset, complex movement disorder - NMDAR
2) Elderly, focal seizures including facio-brachial dystonic seizures, hyponatremia- LGI1
3) Young patients, optic neuritis, transverse myelitis - MOG

Question 7

Partially correct

Mark 3.00 out of 4.00

Theme: The ionic basis of a neuronal action potential
Lead-in: Which statement best describes the defining feature of each phase of an action potential?

1. Refractory period	Voltage-gated potassium channels close and sodium channels open	✗
2. Hyperpolarisation	Membrane potential goes below -70mV	✓
3. Depolarisation	Voltage-gated sodium channels open and sodium flows in	✓
4. Repolarisation	Voltage-gated potassium channels open and sodium channels close	✓

Explanation: An action potential occurs when a local depolarisation to the threshold potential triggers the opening of voltage-gated sodium channels, and the rapid influx of sodium ions brings the membrane potential to a positive value (depolarisation). The membrane potential is restored to its normal resting value (repolarisation) by the delayed opening of the voltage-gated potassium channels and by the closing of the sodium channels. Before the membrane potential returns to a resting state, it becomes hyperpolarised (below -70mV). A refractory period follows an action potential, corresponding to the period when the voltage-gated sodium channels are inactivated.

Ref: Carter M, Essner R, Goldstein N, Iyer M. Guide to Research Techniques in Neuroscience, 3rd Edition. 2022. p. 88.

Correct Answers:
1) Refractory period - Voltage-gated sodium channels are inactivated
2) Hyperpolarisation - Membrane potential goes below -70mV
3) Depolarisation - Voltage-gated sodium channels open and sodium flows in
4) Repolarisation - Voltage-gated potassium channels open and sodium channels close

Question 8

Correct

Mark 1.00 out of 1.00

Which of the following best describes the primary mechanism of action of Cariprazine?

Select one:

- Alpha1 antagonist
- 5HT2A/5HT2C antagonist
- 5HT2A/D2 antagonist
- D2/5HT1A partial agonist ✔
- D3 antagonist

Your answer is correct.

Cariprazine is a D2/5HT1A partial agonist. With its potent 5HT1A partial agonist actions despite lesser 5HT2A antagonism, Cariprazine exhibits a low incidence of drug-induced parkinsonism but some akathisia, which can be greatly reduced by slow-dose titration.

Ref: Carmassi C, et al. A 12-month longitudinal naturalistic follow-up of cariprazine in schizophrenia. Frontiers in Psychiatry. 2024 May 21;15:1382013.

The correct answer is: D2/5HT1A partial agonist

Question 9

Incorrect

Mark 0.00 out of 1.00

Which neurotransmitter is most implicated in the cognitive deficits of schizophrenia?

Select one:

- Glutamate
- Serotonin
- GABA
- Dopamine
- Acetylcholine ✖

Your answer is incorrect.

Dopamine dysfunction has primarily been associated with positive psychotic symptoms. However, striatal dopamine signaling normally has widespread effects on cortical function, and its dysregulation can also contribute to cognitive symptoms. Many researchers now consider cognitive and some negative symptoms of schizophrenia as caused by a deficit of dopamine activity in mesocortical projections to the dorsolateral prefrontal cortex.

Ref: McCutcheon RA et al. Cognitive impairment in schizophrenia: aetiology, pathophysiology, and treatment. Molecular psychiatry. 2023 Jan 23:1-7.Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 95.

The correct answer is: Dopamine

Question 10

Correct

Mark 1.00 out of 1.00

Chlorpromazine is a low-potency antipsychotic, yet it binds with high affinity at targeted receptors in the brain. Which of the following does this produce?

Select one:

- Sedation ✔
- Agitation
- Arousal
- Insomnia
- Rash

Your answer is correct.

Many D2 antagonists have various other pharmacological actions, including anticholinergic, antihistaminic, and alpha-1 antagonist actions, and can combine to contribute to a great deal of sedation by simultaneously blocking several of the neurotransmitters in the arousal pathway. Agents with particularly strong binding at these three receptors, such as chlorpromazine, are sometimes administered when sedation is needed on top of antipsychotic action.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 180-181.

The correct answer is: Sedation

Question 11

Correct

Mark 1.00 out of 1.00

Which of the following is a nicotinic partial agonist?

Select one:

- Varenicline ✔
- Transdermal nicotine
- Acetylcholine
- Acamprosate
- Bupropion

Your answer is correct.

Varenicline is a nicotinic partial agonist. Nicotinic full agonists include acetylcholine, which is very short-acting, and nicotine, which is very long-acting. Bupropion is a norepinephrine-dopamine reuptake inhibitor.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 552-553.

The correct answer is: Varenicline

Question 12

Correct

Mark 1.00 out of 1.00

Which of the following antidepressants is least associated with sexual dysfunction?

Select one:

- Vortioxetine ✔
- Trazodone
- Fluoxetine
- Amitriptyline
- Imipramine

Your answer is correct.

Of the antidepressants, agomelatine, bupropion, and vortioxetine have the best evidence supporting a more favourable sexual side effect profile. Among the tricyclic antidepressants, sexual dysfunction is more common with clomipramine, amitriptyline, and imipramine. Overall evidence suggests relatively high rates of sexual dysfunction across all the SSRIs.

Ref: Montejo, A.L., et al., 2024. Switching to Vortioxetine in Patients with Poorly Tolerated Antidepressant-Related Sexual Dysfunction in Clinical Practice: A 3-Month Prospective Real-Life Study. Journal of Clinical Medicine, 13(2), p.546.

The correct answer is: Vortioxetine

Question 13

Correct

Mark 1.00 out of 1.00

Mirtazapine's actions on which receptor prevent nausea and vomiting?

Select one:

- 5HT2C
- Alpha-2
- H1
- 5HT2A
- 5HT3 ✔

Your answer is correct.

The 5HT3 receptors best known to clinicians are perhaps those localised in the chemoreceptor trigger zone of the brainstem, where they mediate nausea and vomiting (especially in response to cancer chemotherapy), and those localised in the GI tract, where they mediate nausea, vomiting, and diarrhoea/bowel mobility when stimulated by serotonin. Blockage of 5HT3 receptors by mirtazapine can, therefore, protect against chemotherapy-induced nausea and vomiting and against serotonin-induced GI side effects induced by agents that increase serotonin.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 309-311.

The correct answer is: 5HT3

Question 14

Correct

Mark 1.00 out of 1.00

Which of the following is present in neonatal vision?

Select one:

- Convergence
- Color perception
- Accommodation
- Focus on distant objects
- Blink reflex ✔

Your answer is correct.

The fundamental neural circuitry underlying the blink reflex is fully developed at birth. Reflexive contraction of eye's orbicularis oculi muscle in response to direct trigeminal nerve stimulation has been documented as early as 10 to 10.5 weeks of gestation (Humphrey, 1964, as references in Lawrenson et al., 2005).

Ref: Lawrenson JG et al. Tear-film lipid layer morphology and corneal sensation in the development of blinking in neonates and infants. Journal of Anatomy. 2005 Mar;206(3):265-70.
Fanning BA. NEUROLOGIC ASSESSMENT. Tappero and Honeyfield's Physical Assessment of the Newborn: A Comprehensive Approach to the Art of Physical Examination. 2024 Jan 24:181.

The correct answer is: Blink reflex

Question 15

Correct

Mark 1.00 out of 1.00

According to Piaget, at which stage of cognitive development is conservation usually achieved?

Select one:

- Formal operational
- Sensorimotor
- Pre-operational
- Concrete operational ✔
- Post-operational

Your answer is correct.

The stage of concrete operations (7-11 years of age) is so named because, during this period, children operate and act on the concrete, real, and perceivable world of objects and events. Conservation is the ability to recognise that, although the shape of objects may change, they still maintain or conserve other characteristics that enable them to be recognised as the same.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1008-1009.

The correct answer is: Concrete operational

Question 16

Correct

Mark 1.00 out of 1.00

According to Maslow, which need comes after a person has achieved belonging and love?

Select one:

- Transcendence
- Safety
- Self-actualisation
- Esteem ✔
- Physiological

Your answer is correct.

A leader in humanistic psychology, Maslow described a hierarchical organisation of needs present in everyone. As physiological needs (such as hunger and thirst) and safety needs are satisfied, more advanced needs (belonging & love, then esteem) become the primary motivators. Self-actualisation may be considered the highest need, and self-actualisers may also experience a brief powerful transcendent state of consciousness – a sense of heightened understanding, an intense euphoria, an integrated nature, unity with the universe, and an altered perception of time and space – that can produce long-lasting beneficial effects.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1084.

The correct answer is: Esteem

Question 17

Correct

Mark 1.00 out of 1.00

From a very young age, Tom has wanted to be a doctor, just like his father. Now that Tom is a young adult, he is totally committed to being a physician without ever having explored different alternatives. According to Marcia, what is Tom's identity status?

Select one:

- Diffusion
- Achievement
- Moratorium
- Foreclosure ✔
- Confusion

Your answer is correct.

Building on Erik Erikson's work, Canadian psychologist James Marcia identified two core components of identity formation: exploration (actively considering various options) and commitment (making definitive life choices). Based on these, he proposed four identity statuses: 1. Achievement: Individuals have explored different paths and made committed choices. 2. Foreclosure: Commitment to a path without prior exploration. 3. Moratorium: Actively exploring options without making firm commitments. 4. Diffusion: Lack of both exploration and commitment.

Ref: Verschueren M et al. Identity statuses throughout adolescence and emerging adulthood: A large-scale study into gender, age, and contextual differences. Psychologica Belgica. 2017;57(1):32.

The correct answer is: Foreclosure

Question 18

Incorrect

Mark 0.00 out of 1.00

The defence mechanisms of Intellectualisation and rationalisation may be seen in which Piagetian stage?

Select one:

- Sensorimotor
- Pre-operational ✖
- Concrete operational
- Formal operational
- Post-operational

Your answer is incorrect.

Defences can be grouped hierarchically according to the degree of maturity associated with them. Narcissistic defences are the most primitive and may occur in young children. Immature defences are seen in early childhood and pre-adolescent years. Neurotic defences are seen in older children/adolescents and in apparently normal and healthy adults, often under stress. Mature defences are seen in functioning adults. Intellectualisation and rationalisation are neurotic defences. Piaget's formal operational stage takes place from age 11 until the end of adolescence. Note that Piaget did not discuss psychodynamic defenses within a maturational calendar, but this association was made by subsequent theorists.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 891-893.
Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1008-1009, 1065-1067.

The correct answer is: Formal operational

Question 19

Incorrect

Mark 0.00 out of 1.00

In which Eriksonian stage is a mid-life crisis most likely to occur?

Select one:

- Intimacy vs Isolation
- Industry vs Inferiority
- Identity vs Role Confusion
- Generativity vs Stagnation
- Integrity vs Despair ✖

Your answer is incorrect.

From about ages 40-60 years, the failure of generativity can lead to profound personal stagnation, sometimes masked by various escapisms (such as alcohol and drug abuse and sexual and other infidelities). Mid-life crisis or premature invalidism (physical and psychological) can occur.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1075.

The correct answer is: Generativity vs Stagnation

Question 20

Correct

Mark 1.00 out of 1.00

Which of the following is a specialised form of learning occurring early in life during a particular period referred to as the critical period?

Select one:

- Consummatory response
- Appetitive behaviour
- Imprinting ✔
- Fixed action pattern
- Ritualization

Your answer is correct.

Imprinting is a specialised form of learning that occurs early in life and often influences behaviour later in life. The exposure to the stimulus situation must occur during a particular period, the critical period and the exposure can be short-duration and without obvious reward. Konrad Lorenz, an ethologist involved in the systematic study of animal behaviour, is best known for his studies of imprinting.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1042.

The correct answer is: Imprinting

Question 21

Correct

Mark 1.00 out of 1.00

The ability to delay gratification in childhood is associated with

Select one:

- Oppositional defiant disorder
- Conduct disorder
- Long-term academic performance ✔
- Low intelligence
- Social and emotional difficulties

Your answer is correct.

The capacity for delay of gratification in children has been correlated with a range of positive long-term outcomes, including enhanced social responsibility, competence, and improved academic achievement. Some studies also link this to higher intelligence. As highlighted by Twito et al (2019), the delayed gratification capacity has been recognized as a protective factor against a range of psychological and physical health issues, such as conduct disorders, antisocial behaviors, hyperactivity, addictive behaviors, and obesity.

Ref: Twito L et al. The motivational aspect of children's delayed gratification: Values and decision making in middle childhood. Frontiers in psychology. 2019 Jul 31;10:386656

The correct answer is: Long-term academic performance

Question 22

Correct

Mark 1.00 out of 1.00

When experienced in childhood, which of the following has been linked to an increased risk of future delinquency?

Select one:

- Early childhood viral illness
- Parental criminality/incarceration ✔
- Immigration
- Being adopted
- Physical injury

Your answer is correct.

Adverse childhood experiences (ACEs) have been described as 'potentially traumatic events that occur in childhood (0-17 years).' They include sexual, physical, and emotional abuse; witnessing violence at home; living with someone who has a mental illness; parental separation or divorce; parental incarceration; and household alcohol or illicit drug problems. ACEs have been linked to a higher probability of future criminal behaviour.

Ref: Cinini SF, Mkhize SM. Criminal Behavior and Youth Crime: A Juvenile Delinquency Perspective on Adverse Childhood Experience. InCriminal Behavior-The Underlyings, and Contemporary Applications 2023. IntechOpen.

The correct answer is: Parental criminality/incarceration

Question 23

Correct

Mark 1.00 out of 1.00

Individuals with callous-unemotional traits in conduct disorder tend to display

Select one:

- High reactivity towards stress
- Preference for novelty-seeking ✔
- Little to no aggression
- High levels of guilt
- High effort in school

Your answer is correct.

Conduct disorder can be subtyped according to the presence or absence of callous-unemotional (CU) traits (deficits in empathy and guilt). Those with CU traits tend to seek out novel and dangerous activities and show less physiological arousal to negative emotional stimuli. Other indicators of the CU construct include the absence of effort to do well in important activities and constricted or superficial displays of emotion.

Ref: Fairchild G et al. Conduct disorder. Nature Reviews Disease Primers. 2019 Jun 27;5(1):43.
Frick PJ, Kemp EC. Conduct disorders and empathy development. Annual Review of Clinical Psychology. 2021 May 7;17:391-416.

The correct answer is: Preference for novelty-seeking

Question 24

Correct

Mark 1.00 out of 1.00

When is performance suboptimal in the classic 'Yerkes-Dodson arousal curve'?

Select one:

- Throughout the curve
- Arousal too high
- Arousal too low
- Arousal at a mid-point
- Arousal too high or too low ✔

Your answer is correct.

The classic Yerkes-Dodson law describes a curvilinear relationship between arousal level and cognitive performance. Performance is suboptimal in the classic Yerkes-Dodson arousal curve if the arousal is too high or too low. Too little arousal can lead to reduced motivation levels, and too much arousal can lead to a stressful reaction affecting performance. Hebb also proposed an inverted U-shaped relationship between arousal and performance.

Ref: Nieuwenhuis S. Arousal and performance: revisiting the famous inverted-U-shaped curve. Trends in Cognitive Sciences. 2024 Apr

The correct answer is: Arousal too high or too low

Question 25

Correct

Mark 1.00 out of 1.00

Which neurocognitive measure has the most negative association with ADHD polygenic scores?

Select one:

- Spatial reasoning
- Non-verbal reasoning
- Verbal reasoning
- Working memory
- Attention ✔

Your answer is correct.

The negative association of ADHD risk variants with executive functions, especially attention, is in line with the inattention problems often observed in individuals with ADHD. Other neurocognitive measures negatively associated with ADHD polygenic scores include working memory, verbal reasoning, non-verbal reasoning, spatial reasoning, and facial memory.

Ref: Demontis D et al. Genome-wide analyses of attention deficit hyperactivity disorder identify 27 risk loci, refine the genetic architecture, and implicate several cognitive domains. Nature genetics. 2023 Feb;55(2):198.

The correct answer is: Attention

Question 26

Correct

Mark 1.00 out of 1.00

The propensity to change one's behaviour due to being observed is known as the

Select one:

- Rosenthal effect
- Genovese effect
- Hawthorne effect ✔
- Pygmalion effect
- Bystander effect

Your answer is correct.

The propensity to change one's behaviour due to being observed is known as the Hawthorne effect. This notion originated from studies that were later discredited for methodological flaws. Despite its questionable validity, this concept remains influential across various disciplines where human behaviour is studied.

Ref: Vannan K. History of the Hawthorne Effect. The Encyclopedia of Research Methods in Criminology and Criminal Justice. 2021 Aug 24;1:246-8.

The correct answer is: Hawthorne effect

Question 27

Correct

Mark 1.00 out of 1.00

Using his imagination, a patient is exposed to a phobic object for a long period of time without his usual means of escape. What is this technique called?

Select one:

- Systemic desensitisation
- Flooding
- Implosion ✔
- Aversion therapy
- Covert sensitisation

Your answer is correct.

In behaviour therapy, it is thought that high levels of anxiety cannot be maintained for long periods, and a process of 'exhaustion' occurs. By exposing a patient to a phobic object and preventing their usual escape or avoidance, there is the extinction of the usual, maladaptive anxiety response. This may be done in vivo (called 'flooding') or in imagination (called 'implosion').

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 908.

The correct answer is: Implosion

Question 28

Incorrect

Mark 0.00 out of 1.00

Which among the following is provided by Thurstone scale?

Select one:

- Qualitative data regarding attitudes
- Quantitative data regarding attitudes
- Qualitative data regarding behaviour ✖
- Quantitative data regarding behaviour
- Qualitative and quantitative data regarding behaviour and attitudes

Your answer is incorrect.

The Thurstone scale consists of statements about a particular issue, each with a numerical value indicating the respondent's attitude about the issue. Thurstone scaling can be an attractive methodology for arriving at quantitative measures. However, developing a true Thurstone scale is considerably harder than describing one, and the practical problems associated with the method often outweigh its advantages unless the researcher has a compelling reason for wanting the type of calibration that it provides.

Ref: DeVellis RF, Thorpe CT. Scale development: Theory and applications. Sage publications; 2021.

The correct answer is: Quantitative data regarding attitudes

Question 29

Correct

Mark 1.00 out of 1.00

At the start of the school year, a teacher tells his students that he anticipates that they will get good grades if they use school-issued pencils. Indeed, all the students who use these pencils get very good grades. This demonstrates which of the following?

Select one:

- Bystander effect
- Rosenthal effect ✔
- Halo effect
- Domino effect
- Hawthorne effect

Your answer is correct.

Rosenthal's doctoral research, rooted in the Thematic Apperception Test (TAT), contrasted projection patterns between people with schizophrenia and a healthy control group following an experimental manipulation. A surprising finding emerged from the dissertation's preliminary data: the groups already exhibited differences aligned with Rosenthal's hypothesized outcome. This unexpected discovery marked the inception of the experimenter expectancy effect, often termed the "Rosenthal effect." The Rosenthal effect is a particular type of experimenter effect in which the experimenter's expectations of the likely outcome of the experiment act as a self-fulfilling prophecy, biasing the results in the direction of the expectation. This is very similar to the Pygmalion effect, in which people tend to behave the way others expect them to.

Ref: Friedman, H. S. (2024). Robert Rosenthal (1933–2024). American Psychologist. Advance online publication. https://doi.org/10.1037/amp0001356

The correct answer is: Rosenthal effect

Question 30

Correct

Mark 1.00 out of 1.00

Which of the following theories explains why there are higher rates of severe mental illness in urban neighbourhoods characterised by high levels of deprivation, poverty, and ethnic heterogeneity?

Select one:

- Social drift ✔
- Double bind
- Schizophrenogenic mother
- Dopamine theory
- Marital skew

Your answer is correct.

According to the social drift theory, the symptoms and deterioration of cognitive functioning associated with severe mental illness lead to increasing difficulties in function and maintenance of living standards. Thus, individuals drift progressively into lower socioeconomic status or more deprived areas.

Ref: Lee SC et al. Area deprivation, urbanicity, severe mental illness and social drift—A population-based linkage study using routinely collected primary and secondary care data. Schizophrenia Research. 2020 Jun 1;220:130-40.

Kirkbride JB, et al. The social determinants of mental health and disorder: evidence, prevention and recommendations. World psychiatry. 2024 Feb;23(1):58.

The correct answer is: Social drift

Question 31

Incorrect

Mark 0.00 out of 1.00

Which of the following is true of musical hallucinations?

Select one:

- Do not exist; all auditory hallucinations are verbal
- More commonly occur in current/former musicians ✖
- Are associated with a rudimentary tone
- Seldom involve familiar melodies
- Are associated with auditory impairment/deafness

Your answer is incorrect.

Not all auditory hallucinations are verbal in nature. Musical hallucinations are reported in individuals with deafness. They are often thought of as the auditory equivalence of Charles Bonnet Syndrome, a condition that is associated with visual hallucinations in the context of visual impairment. Musical hallucinations often involve familiar melodies such as ballads and hymns.

Ref: Oyebode F. Sims' symptoms in the mind: Textbook of descriptive psychopathology. 7th edition, 2022. p. 98.

The correct answer is: Are associated with auditory impairment/deafness

Question 32

Incorrect

Mark 0.00 out of 1.00

A measurement instrument utilising pain scores from 'no pain' to 'excruciating pain' is considered which type of scale?

Select one:

- First rank
- Thurstone
- Likert
- Semantic differential ✖
- Qualitative

Your answer is incorrect.

A Likert scale is a very commonly used self-report measurement instrument that provides a convenient way to measure unobservable constructs.

Ref: Jebb AT, et al. A review of key Likert scale development advances: 1995–2019. Frontiers in psychology. 2021 May 4;12:637547.

The correct answer is: Likert

Question 33

Correct

Mark 1.00 out of 1.00

The two-factor theory of delusion formation includes which of the following?

Select one:

- Meta-representations and theory of mind
- Neuropsychological deficit, anomalous experience, and biased cognitive process ✔
- Abnormal neuronal migration and the DISC1 gene
- Excessive synaptic pruning and the C4 gene
- Dopaminergic overactivity, glutaminergic hypoactivity, and serotonergic overactivity

Your answer is correct.

The two-factor theory of delusion formation is very influential. It proposes that at least two factors cause monothematic delusions. Factor 1 is a neuropsychological deficit responsible for the anomalous data that may result in an anomalous experience (e.g., a perceptual defect in skin conductance response on seeing familiar faces). Factor 2 is a cognitive process (described as either dysfunctional or biased belief formation) explaining either the initial factor with a delusional interpretation or the prolonged maintenance of the delusional belief in the face of mounting counterevidence. Factor 2 is required to explain the transition from the data resulting in an anomalous experience to the delusional belief. Some theorists do not consider the first factor as sufficiently specific (Corlett, 2019).

Ref: Lancellotta E, Bortolotti L. Delusions in the two-factor theory: pathological or adaptive?. European Journal of Analytic Philosophy. 2020 Nov 8;16(2):37-57.
Corlett PR. Factor one, familiarity and frontal cortex: A challenge to the two-factor theory of delusions. Cognitive Neuropsychiatry. 2019 May 4;24(3):165-77

The correct answer is: Neuropsychological deficit, anomalous experience, and biased cognitive process

Question 34

Incorrect

Mark 0.00 out of 1.00

A man is persecuted by the government in his home country. After finding asylum in a new country, he continues to think that he is under surveillance through closed-circuit television (CCTV). As a result, he constantly feels unsettled. This is best understood as which of the following?

Select one:

- Persecutory delusion ✖
- Overvalued idea
- Idea of reference
- Sensitive paranoia
- Delusion of control

Your answer is incorrect.

An overvalued idea is an acceptable, comprehensible idea pursued by the individual beyond the bounds of reason. This is a thought that, because of the associated feeling tone, takes precedence over all other ideas and maintains this precedence permanently or for a long period of time. Overvalued ideas tend to be less fixed than delusions and usually have some degree of basis in reality. Compared to overvalued ideas, delusions tend to be characterised by abrupt onset, implausible content, and a relative indifference to other people's opinions.

Ref: Oyebode F. Sims' symptoms in the mind: Textbook of descriptive psychopathology. 7th edition, 2022. p. 131.
Casey P, Kelly B. Fish's clinical psychopathology: Signs and symptoms in psychiatry, 5th Edition. 2024. p. 48.

The correct answer is: Overvalued idea

Question 35

Correct

Mark 1.00 out of 1.00

Which of the following is true of phantom limb pain?

Select one:

- These phenomena are restricted to limb injuries
- Pain is felt as though it is inside an amputated limb ✔
- It usually begins years after a traumatic amputation
- It is more common if there is no pain prior to the amputation
- Pain is at the site of the amputation/surgical incision

Your answer is correct.

Phantom limb pain is felt as though it were in an amputated limb. It usually begins soon after an amputation and has a self-limited duration of several weeks. The pain is more likely to develop and persist if the affected body part was chronically painful before the amputation (as with osteomyelitis). The phenomena are not restricted to limb injuries. They may also occur after other procedures such as mastectomy, enucleation of the eye, removal of the larynx, or the construction of a colostomy. Pain at the site of the amputation/surgical incision, called ‘stump pain’, differs from phantom limb pain because it is confined to the injury and is attributable to nerve scars.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman’s Clinical Neurology for Psychiatrists, 9th Edition. 2023. pp. 335-336.
Casey P, Kelly B. Fish’s clinical psychopathology: Signs and symptoms in psychiatry, 5th Edition. 2024. pp.36-37.

The correct answer is: Pain is felt as though it is inside an amputated limb

Question 36

Correct

Mark 1.00 out of 1.00

Which of the following features is more suggestive of complex PTSD than borderline personality disorder?

Select one:

- Difficulties in relationships due to idealisation and devaluation
- Persistent negative sense of self ✔
- Vacillation between the positive and negative sense of self
- Repeated efforts to connect with others to avoid feelings of abandonment
- Extreme strategies to regulate affect

Your answer is correct.

While borderline personality disorder (BPD) and complex PTSD (CPTSD) exhibit overlap in the type of difficulties across affect regulation, self-concept, and interpersonal relationships, there are also important distinctions. For example, in CPTSD, there is a persistent negative sense of self, while in BPD there is an unstable sense of self that can be internalising or positive and may change back and forth between the two. Interpersonal difficulties in CPTSD are often characterised by avoidance and disconnection, while in BPD, they may include relationships marked by either ongoing or intermittent volatility and by efforts to connect with others to avoid feelings of abandonment. BPD is also marked by more extreme strategies to regulate affect.

Ref: Powers A et al. Distinguishing PTSD, complex PTSD, and borderline personality disorder using exploratory structural equation modeling in a trauma-exposed urban sample. Journal of anxiety disorders. 2022 May 1;88:102558.

The correct answer is: Persistent negative sense of self

Question 37

Incorrect

Mark 0.00 out of 1.00

Which of the following is considered a first-rank symptom of schizophrenia?

Select one:

- Thought insertion
- Thought blocking
- Command hallucinations
- Flight of ideas
- Paranoid delusions ✖

Your answer is incorrect.

There are eleven first-rank symptoms (of special significance, but not pathognomonic) of schizophrenia, organised into four categories according to type:

Auditory hallucinations: 1. voices heard arguing, 2. thought echo, 3. running commentary
Delusions of thought interference: 4. thought insertion, 5. thought withdrawal, 6. thought broadcasting
Delusions of control: 7. passivity of affect, 8. passivity of impulse, 9. passivity of volitions, 10. somatic passivity
11. Delusional perception: a primary delusion of any content that is reported by the patient as having arisen following the experience of a normal perception

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 78.

The correct answer is: Thought insertion

Question 38

Correct

Mark 1.00 out of 1.00

What is the most common target in passive immunotherapy for Alzheimer's disease?

Select one:

- Inflammatory regulators
- Tau
- NMDA receptors
- Alpha synuclein
- Amyloid beta ✔

Your answer is correct.

Based on the amyloid cascade hypothesis, the accumulation of amyloid beta peptides and the deposition of amyloid plaques is considered the main trigger of the neurodegenerative process of Alzheimer disease (AD), although controversy remains. Amyloid beta has long been the major focus of efforts to develop passive immunotherapy for AD. Passive immunotherapy targeting tau and neuroinflammation is at an early stage of development.

Ref: Guo X et al. Passive immunotherapy for Alzheimer's disease. Ageing Research Reviews. 2024 (94): 1-16

The correct answer is: Amyloid beta

Question 39

Correct

Mark 1.00 out of 1.00

Which of the following is usually the most important investigation to diagnose autoimmune encephalitis?

Select one:

- Brain biopsy
- Brain MRI
- EEG
- CSF autoantibodies ✔
- Serum autoantibodies

Your answer is correct.

Autoimmune encephalitis (AIE) comprises a group of non-infectious immune-mediated inflammatory disorders in the brain parenchyma, often involving the cortical or deep grey matter with or without the involvement of the white matter, meninges, or spinal cord. In addition to ruling out alternative diagnoses, standard brain MRI with contrast can show changes consistent with one or more of the AE anatomical syndromes, though MRI may also be normal. CSF analysis is the most important test in AE evaluation and is usually the second step in the workup after brain MRI. AIE with NMDAR antibodies is the most frequent AIE subtype at younger ages and is almost always associated with inflammatory CSF findings. In contrast, anti-LGI1 AIE is the most frequent AIE subtype in the elderly, and in most patients, CSF is normal.

EEG is commonly performed to exclude subclinical status epilepticus in encephalopathic patients or to monitor treatment response in patients with seizures. Most AE cases with normal brain MRI or typical MRI patterns do not require a brain biopsy. Rarely, a brain biopsy may be needed for atypical or mass-like lesions to exclude neoplastic or other possibilities, especially when all other investigations point away from autoimmunity.

Ref: Abboud H et al. Autoimmune encephalitis: proposed best practice recommendations for diagnosis and acute management. Journal of Neurology, Neurosurgery & Psychiatry. 2021 Jul 1;92(7):757-68.
Cerebrospinal Fluid Findings in Patients With Autoimmune Encephalitis—A Systematic Analysis, Tetyana Blinder and Jan Lewerenz

The correct answer is: CSF autoantibodies

Question 40

Incorrect

Mark 0.00 out of 1.00

Neuroimaging studies in bipolar disorder have found which of the following?

Select one:

- Widespread increases in cortical thickness
- Lower fractional anisotropy with short disease duration
- Higher fractional anisotropy with lithium medication
- Higher fractional anisotropy in unmedicated individuals ✖
- Specific and isolated brain abnormalities

Your answer is incorrect.

Diffusion tensor imaging (DTI) studies, which enable in vivo investigation of structural brain connectivity, have consistently revealed aberrant brain structure in bipolar disorder. Meta-analyses of whole-brain DTI data have demonstrated reduced fractional anisotropy (FA) in people with bipolar disorder. FA, a metric indicative of white matter fiber tract integrity, was inversely correlated with disease severity. Moreover, higher FA values in specific brain regions have been linked to factors associated with better prognosis, such as lithium treatment, later disease onset, and shorter illness duration.

Ref: Al-Sharif NB, Zavaliangos-Petropulu A, Narr KL. A review of diffusion MRI in mood disorders: mechanisms and predictors of treatment response. Neuropsychopharmacology. 2024 Jun 20:1-9.

The correct answer is: Higher fractional anisotropy with lithium medication

Question 41

Correct

Mark 1.00 out of 1.00

Where in the brain is the preoptic area essential in thermoregulation?

Select one:

- Thalamus
- Pons
- Hypothalamus ✔
- Cerebellum
- Dorsolateral prefrontal cortex

Your answer is correct.

The preoptic area of the hypothalamus has been demonstrated to be a key brain region that integrates peripheral and central temperature information to control body temperature.

Ref: Rothhaas R, Chung S. Role of the preoptic area in sleep and thermoregulation. Frontiers in Neuroscience. 2021 Jul 1;15:664781.

The correct answer is: Hypothalamus

Question 42

Correct

Mark 1.00 out of 1.00

The dual orexin receptor antagonist (DORA) suvorexant treats which of the following?

Select one:

- ADHD
- Mania
- Insomnia ✔
- Psychosis
- Tremor

Your answer is correct.

Dual orexin receptor antagonists (DORAs) block the wake-stabilising effects of orexins, especially at orexin 2 receptors. After administration of a DORA, arousal is no longer enhanced, and orexins no longer stabilise wakefulness, and the patient goes to sleep. Both suvorexant and lemborexant improve not only the initiation but also the maintenance of sleep.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 423.

The correct answer is: Insomnia

Question 43

Correct

Mark 1.00 out of 1.00

Hebbian associative synaptic plasticity in the form of long-term potentiation is well-established in neuroscience as the primary neural mechanism behind

Select one:

- Colour perception
- Sleep
- Long-term memory ✔
- Motor speed
- Delirium

Your answer is correct.

Hebbian associative synaptic plasticity in the form of long-term potentiation is well-established in neuroscience as the primary neural mechanism behind long-term memory.

Ref: Lansner A et al. Fast Hebbian plasticity and working memory. Current opinion in neurobiology. 2023 Dec 1;83:102809.

The correct answer is: Long-term memory

Question 44

Correct

Mark 1.00 out of 1.00

Depersonalisation is most likely a phenomenon produced by which brain region?

Select one:

- Insula
- Anterior temporal region
- Multiple neural pathways ✔
- Amygdala
- Occipital lobe

Your answer is correct.

It is unlikely that any one brain region or mechanism alone can fully explain depersonalisation. Instead, many neural pathways together likely contribute to its pathogenesis. Depersonalisation is regularly cited as a common symptom associated with organic states, especially temporal lobe epilepsy. It has also been associated with increased prefrontal activation and reduced activation in insula/limbic-related areas to aversive, arousing emotional stimuli.

Ref: Murphy RJ. Depersonalization/derealization disorder and neural correlates of trauma-related pathology: a critical review. Innovations in Clinical Neuroscience. 2023 Jan;20(1-3):53.
Oyebode F. Sims' symptoms in the mind: Textbook of descriptive psychopathology. 7th edition, 2022. pp. 202-203.
Casey P, Kelly B. Fish's clinical psychopathology: Signs and symptoms in psychiatry, 5th Edition. 2024. p. 81.

The correct answer is: Multiple neural pathways

Question 45

Incorrect

Mark 0.00 out of 1.00

Which of the following has been shown to be decreased in the brain following ECT?

Select one:

- Serotonin
- N-acetyl aspartate
- BDNF ✖
- Dopamine
- Acetylcholine

Your answer is incorrect.

There is strong evidence to support a reduction in N-acetyl aspartate (NAA), considered a marker of neuronal integrity, following ECT. Several studies have found lower NAA levels after ECT treatment.

Ref: Erchinger VJ et al. Magnetic resonance spectroscopy in depressed subjects treated with electroconvulsive therapy—a systematic review of literature. Frontiers in psychiatry. 2021 Mar 25;12:608857.
Erchinger VJ et al. Electroconvulsive therapy triggers a reversible decrease in brain N-acetylaspartate. Frontiers in Psychiatry. 2023 Jun 9;14:1155689.

The correct answer is: N-acetyl aspartate

Question 46

Correct

Mark 1.00 out of 1.00

Of the following, which brain region is most implicated in producing impulsivity in mania?

Select one:

- Insula
- Orbitofrontal cortex ✔
- Cerebellum
- Hypothalamus
- Posterior pituitary

Your answer is correct.

In studies of affective and cognitive features of mania, which include response inhibition, impulsivity, and hedonic state, the prefrontal cortex, namely the orbitofrontal cortex (OFC) and anterior cingulate cortex, as well as the amygdala, have been consistently implicated. Some findings suggest that the OFC is involved in evaluating the incentive value of outcomes and disrupting/modulating its activity is associated with impaired response inhibition and impulsive behaviour.

Ref: Cotovio G, Oliveira-Maia AJ. Functional neuroanatomy of mania. Translational Psychiatry. 2022 Jan 24;12(1):29.

The correct answer is: Orbitofrontal cortex

Question 47

Correct

Mark 1.00 out of 1.00

How many primary emotions in the mammalian brain are identified by Panksepp?

Select one:

- Seven ✔
- Five
- Four
- Eleven
- Nine

Your answer is correct.

Mainly using electrical stimulation of the brain, but also pharmacological manipulations and localised brain lesions, Panksepp identified seven emotional brain systems in the mammalian brain: seeking, rage/anger, fear, lust, care, panic/sadness, and play.

Ref: Davis KL, Montag C. Selected principles of Pankseppian affective neuroscience. Frontiers in neuroscience. 2019 Jan 17;12:427897.

The correct answer is: Seven

Question 48

Correct

Mark 1.00 out of 1.00

Which of the following is true about amyloid-related imaging abnormalities with edema (ARIA-E) after passive immunisation for Alzheimer's disease?

Select one:

- There is a higher risk in ApoE4 carriers ✔
- It is not dose-dependent
- It has only occurred in one clinical trial
- It is a rare adverse event
- It is almost always fatal

Your answer is correct.

Amyloid-related imaging abnormalities with edema or effusion (ARIA-E) is a vasogenic effect seen in multiple clinical trials of passive immunotherapy for Alzheimer's disease and is a common adverse event. Many cases are asymptomatic, though patients with serious ARIA-E events have died. ARIA-E is often dose-dependent and seen more commonly in ApoE4 carriers. To decrease the risk of ARIA-E, gradual up-titration schemes have been used. High-dose and low-dose groups have also been designed based on patients carrying the ApoE4 gene.

Ref: Guo X et al. Passive immunotherapy for Alzheimer's disease. Ageing Research Reviews. 2024 (94): 1-16.

The correct answer is: There is a higher risk in ApoE4 carriers

Question 49

Incorrect

Mark 0.00 out of 1.00

Visual hallucinations are understood to be produced in which part of the brain?

Select one:

- Vision-related networks
- Lingual gyrus
- Primary visual cortex ✖
- Superior temporal region
- Hypothalamus

Your answer is incorrect.

Uncertainties remain whether there is a single underlying mechanism for visual hallucinations (VH), or whether they have different disease-dependent causes. The functional and structural changes differ between studies of VH, both within a given condition and across conditions, but likely all form part of a distributed network. Pathology involving any part of the network can result in dysfunction that leads to VH. In general, VH is associated with diffuse involvement of vision-related networks.

Ref: O'Brien J et al. Visual hallucinations in neurological and ophthalmological disease: pathophysiology and management. Journal of Neurology, Neurosurgery & Psychiatry. 2020 May 1;91(5):512-9.
van Ommen MM et al. Visual hallucinations in psychosis: the curious absence of the primary visual cortex. Schizophrenia bulletin. 2023 Mar 1;49(Supplement_1):S68-81

The correct answer is: Vision-related networks

Question 50

Correct

Mark 1.00 out of 1.00

Which of the following gives a high signal in T2-weighted MRI?

Select one:

- Water ✔
- Calcium
- Fat
- Hemoglobin
- N-acetylaspartate

Your answer is correct.

T2 pulses last four times as long as T1 pulses and the collection times are also extended to emphasise the signal from hydrogen nuclei surrounded by water. Thus, brain tissue is dark, and CSF is white on T2 images. Areas within the brain tissue with abnormally high water content, such as tumours, inflammation, and strokes, appear brighter on T2 images.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 70.

The correct answer is: Water

Question 51

Correct

Mark 1.00 out of 1.00

According to Vygotsky, the distance between a child's actual developmental level and their higher level of potential development is called

Select one:

- Zone of proximal development ✔
- Separation-individuation
- Redirection activity
- Imprinting
- Critical period

Your answer is correct.

According to Vygotsky, the zone of proximal development is the distance between a child's actual developmental level (as determined by independent problem-solving) and their higher level of potential development (determined through problem-solving under adult guidance or in collaboration with more capable peers). Vygotsky believed that this is most likely to emerge during play.

Ref: Ness IJ. Zone of proximal development. In The Palgrave Encyclopedia of the Possible 2023 Jan 26 (pp. 1781-1786). Cham: Springer International Publishing.

The correct answer is: Zone of proximal development

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